

ABC Data Security Plan

Joshua Schwenk

Colorado State University Global

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Dr. Robert Marohn

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Problem Statement

ABC Company has been in a high-growth period over the last fiscal year and plans for further expansion within the next several quarters. However, due to the limitations of their current data environment, it has been increasingly difficult to keep up with the increasing demands. The company works with large amounts of heterogeneous data, much of which contains sensitive information and is crucial for internal/external operations. Due to the exponential data growth and aging infrastructure, ABC's data warehouse can no longer support processing as efficiently as it once did. Additionally, the ongoing cost of maintenance has created additional constraint on internal resources.

In order to improve scalability, strengthen system integration, and reduce long-term maintenance costs, leadership has decided to migrate their data infrastructure into the cloud. Although cloud technologies offer great efficiency and cost savings, they also introduce greater security concerns. These concerns include data protection, access management, compliance, and regulatory accountability. To reduce the risk associated with the cloud, leadership requires a robust data security plan that defines how sensitive information will be protected across the organization. These strong security controls will come with a high initial cost; however, they will help reduce further expenses associated with cloud security risks and inefficient infrastructure. As a result, leadership must prioritize security investments that address the highest risk areas of the data environment and balance implementation costs with operational efficiency.

Goals and Objectives

To ensure ABC Company's security plan is aligned with operational needs and business strategy, it is important for the organization to first establish clear goals and objectives. Setting these priorities early helps leadership confirm that the environment addresses current challenges and supports future growth (Valacich & George, 2020). For ABC, their goals will focus on the protection of sensitive information, supporting scalability and feasibility, establishing accountability, and maintaining data governance.

Goal 1: Protect Sensitive Organizational Data

Protecting sensitive information is the highest priority within ABC Company's data security plan as exposure of this data presents the greatest operational and financial risk. Stair and Reynolds (2018) explain that layered security controls are necessary because sensitive organizational data often moves across multiple systems where exposure risks increase. Security controls must therefore protect sensitive records while still allowing authorized access for operational use.

- **Objective 1.1:** Establish secure storage environments where highly sensitive manufacturing records, vendor agreements, and government contract data are separated according to risk level and protected through controlled storage systems.
- **Objective 1.2:** Implement encryption standards so sensitive information remains protected.

- **Objective 1.3:** Apply password policies requiring strong password requirements and multifactor authentication for systems containing sensitive organizational data.

Goal 2: Build a Security Structure That Supports Scalable & Feasible Data Operations

As ABC Company expands, its security controls must support larger data volumes and more complex system interactions without disrupting operations. Cloud-based environments improve flexibility and scalability by allowing organizations to adjust computing resources as operational demand changes (Al-Sharafi et al., 2023). Therefore, security planning must remain realistic so protection measures strengthen the organization without creating unnecessary operational delays or increasing long-term infrastructure costs.

- **Objective 2.1:** Define how operational data will move through integrated systems and interoperable environments so information remains secure, consistent, and usable as it passes between systems.
- **Objective 2.2:** Develop storage practices that allow lower-risk operational data and higher-risk contractual data to be managed in environments appropriate to their security needs.

Goal 3: Establish Clear Accountability for Data Access & Security Responsibilities

Accountability requires that every user and system interacting with data operates within clearly defined responsibilities. Even when systems expand, ABC Company remains responsible for protecting its information and monitoring how access is granted (National Security Agency, 2024).

- **Objective 3.1:** Establish role-based access controls so employees, managers, and technical staff only access the data necessary for their job responsibilities.
- **Objective 3.2:** Implement security awareness training that teaches employees how to recognize threats such as phishing, misuse of credentials, and improper handling of sensitive records.

Goal 4: Maintain Governance Through Formal Data Classification & Security Policies

Security controls must align with internal governance expectations and external regulatory requirements, especially because ABC Company works with government-sensitive information and external vendors. Formal classification and policy standards help ensure that data protection remains consistent across departments.

- **Objective 4.1:** Create a formal data classification structure that labels information according to sensitivity and assigns security controls based on risk. Data classification allows organizations to assign stronger controls to higher-risk information before it moves across systems (Stair & Reynolds, 2018).
- **Objective 4.2:** Establish security policies that define acceptable data handling practices, access approval requirements, and governance expectations across departments.

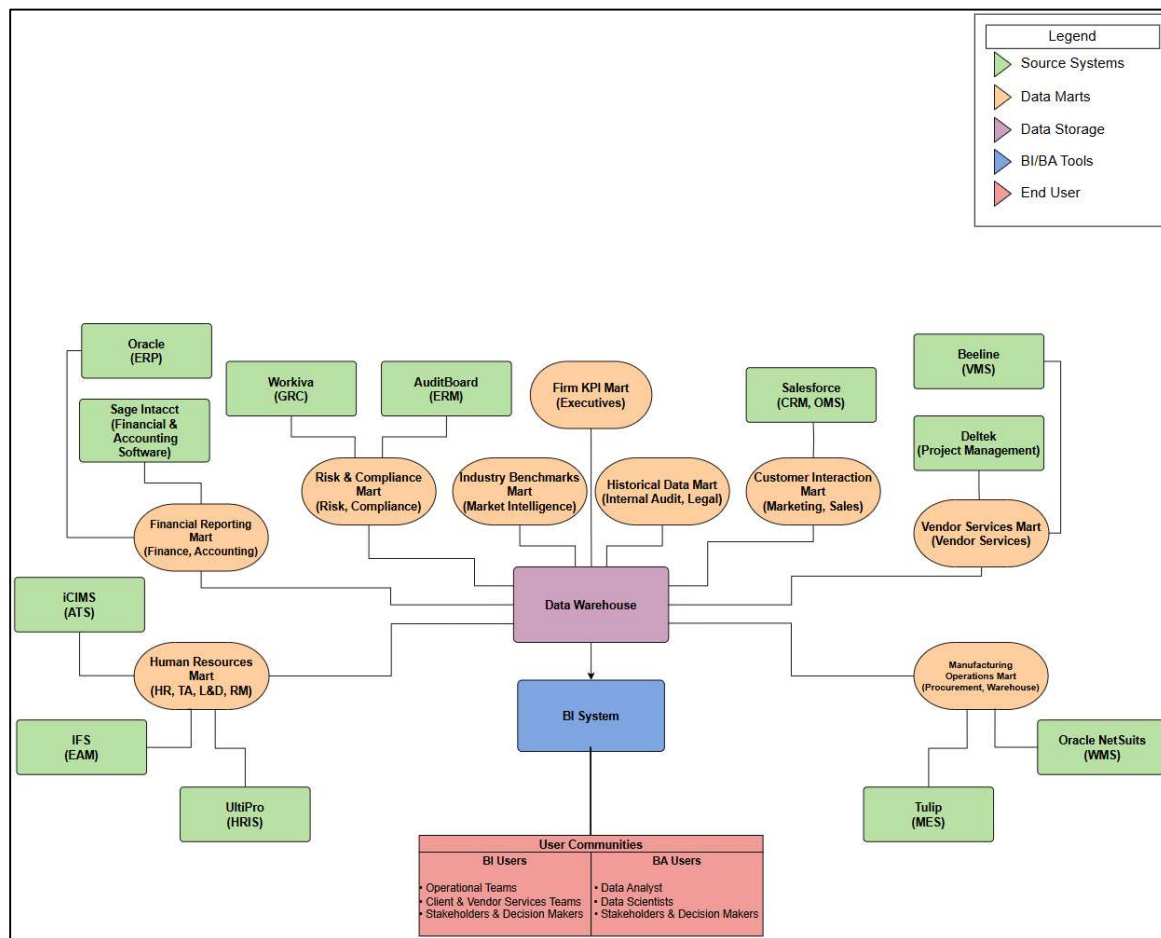
Organizational Assets

ABC Company's organizational assets include the operational systems, administrative platforms, and supporting infrastructure that collectively manages the organization's heterogeneous data environment. As shown in Figure 1, these assets are distributed across multiple business functions before being consolidated into a centralized data warehouse that supports reporting and decision-

making. These assets operate across both internally managed infrastructure and external cloud-supported environments, including internal application servers, centralized storage platforms, and externally hosted systems. The organization relies on enterprise systems that support financial management, human resources, compliance oversight, manufacturing operations, vendor coordination, and customer activity, each of which contributes data with different security and operational requirements. ABC's systems that manage sensitive data (financial records, customer information, employee information, vendor agreements, etc.) represent higher risk assets and should be prioritized for stronger security controls and monitoring. In addition to these business systems, the data warehouse and business intelligence (BI) environment represent critical assets because they aggregate information from across the enterprise and make it accessible to both operational users and decision makers. Because these assets support both daily operations and strategic analysis, they must be protected according to their operational importance and the sensitivity of the information they contain. Beyond security risks, weak data governance within the data warehouse and BI environment can also lead to inaccurate analysis and decision-making, resulting in more operational inefficiencies and higher financial risk.

Figure 1

ABC Company Enterprise Data Environment



Note. Adapted by the author from *Proposal Project: Baker Tilly business intelligence systems architecture* (Schwenk, 2025). Figure was revised to reflect ABC Company's enterprise systems, functional data marts, centralized data storage, reporting tools, and user communities.

Security Policies

Because not all organizational assets contain the same level of risk, ABC Company must apply stronger security controls to systems that store confidential operational and contractual information. When multiple operational systems feed a shared repository, a weakness in one connected system may create a pathway for broader exposure. This can increase the possibility that a single breach could affect multiple parts of the organization's data environment. Because of this, stronger protection is necessary to reduce exposure as one repository increases security risk and the need for carefully controlled access (Aleem & Ahmed, 2015).

One of the primary ways to minimize exposure across connected systems is through role-based access controls that limit how users interact with sensitive information throughout the organization. The National Institute of Standards and Technology (2020) recommends limiting permissions according to operational necessity in order to reduce unnecessary exposure across enterprise systems. Access should be granted according to operational responsibility so employees only view, modify, or transfer data necessary for their assigned duties. Administrative privileges should be restricted to designated technical personnel, with elevated permissions requiring formal approval and periodic review to reduce unnecessary risk. Since the business and data landscape are always evolving, access permissions should be reviewed on an ongoing basis by IT and data owners to ensure access needs remain aligned. Reviews should occur quarterly for systems containing sensitive or high-risk data and annually for lower-risk systems. Additional audits should be conducted following major changes to business processes, role structures, or data procedures, as well as in response to security incidents. The process should be treated as a shared responsibility across IT, data owners, leadership, and end users to maintain accountability and ensure that access controls continue to reflect operational needs. This approach helps ensure that sensitive financial, contractual, and compliance-related information remains available only to individuals whose responsibilities require direct interaction with that data.

Password and encryption policies provide an additional layer of protection by securing access points and protecting data as it moves across ABC Company's systems. Password requirements should include strong standards and multifactor authentication for systems containing sensitive records. These controls help reduce the likelihood that compromised credentials could be used to access multiple connected systems within the enterprise environment. Encryption should also be applied to sensitive data so information remains protected. Stair and Reynolds (2018) explain that encryption reduces exposure by transforming readable information into protected formats that require authorized credentials or keys for access. Together, password and encryption controls strengthen the organization's ability to protect critical data even when systems are integrated across multiple operational functions.

Because security controls must operate across multiple connected systems, ABC Company should implement major policy changes through a phased integration approach rather than applying new controls across all environments at once. Phased implementation allows technical teams to validate

encryption settings, access permissions, and monitoring controls within smaller operational areas before expanding protections across the broader enterprise. This approach also allows implementation timelines to remain manageable by reducing operational disruption and giving technical teams time to correct issues before broader deployment. Staged technology adoption helps organizations adjust computing resources gradually while reducing disruption during system transitions (Al-Sharafi et al., 2023). This approach also makes it easier to identify weaknesses before they affect multiple systems and allows security investments to be distributed over time. In areas where contract integrity or traceability is especially important, selective technologies such as blockchain-based logging may support stronger audit visibility by creating records that are more resistant to unauthorized modification. This approach allows leadership to prioritize higher-risk systems first while managing overall implementation cost and operational impact.

Data Integration, Interoperability, and Classification

Since ABC Company relies on multiple enterprise systems that feed a centralized warehouse, data must remain both integrated and interpretable as it moves across operational environments. Data integration is managing how information moves and is consolidated within and between systems (Gils, 2020). This becomes increasingly important when data originates from separate business functions such as human resources, manufacturing, vendor services, and customer activity. In ABC Company's environment, integration supports scalability by allowing data from these systems to move into a common repository where enterprise reporting can occur without requiring separate analysis within each source system. A centralized structure also helps reduce the likelihood of creating what Gils (2020) describes as a "spaghetti landscape," where too many direct system-to-system connections become difficult to manage and maintain over time. Without shared integration standards, systems may exchange data successfully while still creating inconsistent outputs, which makes interoperability equally important as the organization grows. However, centralizing data also increases the potential impact of a security breach, as more sensitive information is concentrated within a single environment. As a result, maintaining consistent integration standards within centralized systems becomes a priority to reduce security and reporting risk. This reinforces the need for strong access controls and continuous monitoring to limit exposure across interconnected systems.

Interoperability extends this process by ensuring that data remains understandable and usable after it has been integrated across systems (Gils, 2020). For ABC Company, interoperability requires common data definitions and shared standards so information remains consistent. For example, one system may store customer names in a single field while another separates first and last names, or one department may classify vendor activity differently than another. As the organization grows, these shared standards improve scalability by allowing new systems or data sources to be added without creating significant reporting disruptions or structural changes.

A classification model supports interoperability by ensuring that data entering ABC Company's centralized environment is consistently understood and managed according to business purpose and security sensitivity. Within ABC Company, these forms of data exist across manufacturing operations, financial reporting, vendor analysis, and customer activity, where each supports different operational

goals. However, from a security perspective, data must also be classified according to sensitivity so protection requirements remain consistent across connected systems. The National Security Agency (2024) emphasizes that assigning protection levels before data moves across environments helps organizations apply stronger safeguards where exposure creates greater operational risk. ABC Company should therefore apply categories such as public, internal use, confidential, and restricted data, with confidential and restricted classifications assigned to employee records, vendor agreements, financial information, and government contract materials.

To apply these standards consistently, ABC Company must also manage how data is prepared and governed before it enters the cloud-based warehouse. Structured extraction, transformation, and loading processes (ETL) should be used to move information from source systems into the centralized warehouse in a controlled manner. Scheduled ETL cycles also allow leadership to balance reporting availability with controlled update timing, reducing the operational risk of uncontrolled real-time changes. Standardizing data before reporting also reduces the long-term cost of manual correction and repeated reconciliation across departments. The ETL process supports consistency by standardizing formats, correcting conflicting records, and aligning data definitions before information is consolidated for reporting (Valacich & George, 2020). Responsibility for this process should primarily rest with the organization's data engineering and information technology teams, who would oversee system integration and data transformation rules. At the same time, business data owners from each business unit should help validate definitions and reporting expectations so technical consistency remains aligned with operational needs.

Security Awareness & Training

Even well-designed security controls are only effective when employees understand how to apply them within daily operations (Valacich & George, 2020). Employee adoption and security behaviors represent a significant source for security risks. This means that ABC Company should prioritize training to reduce the likelihood of human error leading to data exposure. ABC Company should provide security awareness training that explains how data classifications affect practices across business units. The training should be delivered through a combination of companywide security awareness sessions, targeted role-based workshops, monthly security awareness modules, and phishing simulations. Similar to the role-based training approach, employees who work directly with confidential or restricted information should receive instruction that reflects the systems and records most relevant to their responsibilities (Stair & Reynolds, 2018). A role-based approach helps control training costs by focusing deeper instruction where security exposure is highest rather than applying identical training across all roles. For example, human resources staff may require stronger guidance on employee data handling, while manufacturing leaders and vendor managers should understand how operational records and external data must be protected within connected systems. Training should also include common security risks such as phishing attempts, password misuse, unauthorized data sharing, and improper storage of sensitive records.

Additionally, training should also evolve as technological environments change. New employees should receive foundational security instruction during onboarding, while existing employees should

complete annual refresher training or whenever new policy changes are introduced. Delivering training in recurring intervals helps the organization adjust employee behavior gradually rather than requiring large-scale retraining during major system changes. Technical teams responsible for warehouse management, access permissions, and system integration should receive additional instruction on governance expectations so operational changes do not weaken interoperability or security controls. To ensure employees understand and apply security practices within their daily work, ABC Company will measure their security training effectiveness through assessments and phishing exercises. Ongoing reinforcement helps create shared accountability across departments and reduces the likelihood that procedural mistakes introduce unnecessary risk.

Potential for Data Bias

Even though the ETL processes improve consistency, ABC Company must also recognize the potential for data bias when source systems contain incomplete, outdated, or inconsistently labeled records. Bias can occur when one data source is treated as more reliable than another without verifying whether values remain accurate across systems. According to Putrama and Martinek (2024), combining heterogeneous data from multiple systems can introduce quality distortion when update timing, structures, and record definitions are not aligned before consolidation. For example, vendor or customer records may not always be updated at the same time across all business units. This is especially true with batch integration patterns (such as ETL) where target systems get their updates only a few times per day. This can cause reporting and decision-making to rely on data that is not always up to date. Additional biases can emerge when similar records are classified differently across business units. Depending on how the data is grouped, these classification differences can make certain operational trends appear stronger or weaker (Putrama & Martinek, 2024). Decisions based upon biased data can contribute to greater financial implications, including higher operational costs, weaker vendor decisions, inaccurate production planning, or missed revenue opportunities. To help mitigate this, security teams, IT, and data owners should periodically revalidate ABC Company's transformation rules to reflect current business conditions.

Data Analytics Plan

With the collaboration of the data analytics team, ABC Company can further the framework outlined throughout this plan (ETL processes, formal data classification, security training, etc.) to improve its data security strategy. This is accomplished through reporting and dashboard tools that monitor ABC's systems to ensure they are meeting security expectations. According to Stair and Reynolds (2018), organizations can identify unusual behavior early on with effective monitoring systems. For example, security dashboards can help leadership track failed login attempts, unusual access patterns, repeated permission changes, and abnormal data movement between systems. It will be important for ABC Company to identify any suspicious activity before an isolated situation develops into broader security concerns, as early detection reduces potential financial or reputational exposure. Security personnel, data owners, and senior operational leaders should review access activities periodically to confirm that users still require the permissions assigned to their role. These reviews can

include weekly exception monitoring and monthly trend analysis so that suspicious patterns are identified consistently over time.

Data Management Strategy

“Data management is the development, execution, and supervision of plans, policies, programs, and practices that deliver, control, protect and enhance the value of data and information throughout their lifecycle” (Gils, 2020, p. 26). This strategy is especially important for ABC Company as the organization prepares to transition into the cloud environment. Although the organization has established many foundational governance practices, additional optimization is required to effectively utilize and protect their, their customers’, and their vendors’ data.

Document & Content Management

Currently, ABC operates within a heterogeneous data environment where information is generated and stored across several operational systems. The company has implemented a centralized data warehouse; however, it is not consistently utilized as a primary source for reporting and analysis. Due to the lack of document and content management governance, data is often maintained differently across systems. This has led to variations in how ABC stores and retrieves information. Without a unified structure, employees frequently rely on operational systems rather than the centralized source. Their current approach increases the risk of financial loss resulting from operational inefficiencies and reduced data accuracy. On average, each employee spends approximately five hours of lost productivity per week when having difficulties locating documents (eCOPIER, 2025). Decision-making is also at risk. When records are inaccessible or outdated, or when ABC Company relies on incomplete information from its data warehouse, it can lead to unfounded or wrong decisions. This can result in further financial loss and damaging the organization’s strong reputation for quality. A transition to the cloud may amplify these issues by increasing the scale at which data is accessed and shared across systems. Inconsistencies in how documentation and content is managed can spread more quickly, making it harder to maintain control over sensitive records and increasing the likelihood that decisions are made using unreliable information. To mitigate these risks, ABC must establish stronger document and content management governance controls that standardize how information is created and maintained across systems. This includes reinforcing centralized data usage, implementing consistent documentation standards, introducing ongoing training to ensure employees are following established processes, and yearly audits or during major process changes. With this strategy, ABC can improve data reliability and maintain greater control over its information as it migrates into a cloud environment.

Data Quality

Bringing data into a centralized environment alone does not ensure higher data quality. When multiple data sources are integrated without standard controls, data can become inaccurate, inconsistent, incomplete, or duplicated (Colorado State University Global [CSU], 2026). This can lead to substantial financial risk for ABC Company. According to Krantz and Jonker (2026), “over a quarter of organizations estimate they lose more than USD 5 million annually due to poor data quality, with 7% reporting losses of USD 25 million or more” (para. 2). This makes it necessary for ABC to establish clear processes for maintaining data across its systems. Data quality in this context refers to ensuring that the

information is valid, accurate, complete, uniform, and relevant, all of which must be actively managed as data is created and updated across operational systems (CSU, 2026). Without these controls in place, the risk associated with centralized storage can persist. Improving data quality will require ongoing coordination between governance, IT, and data owners. The teams will define data standards that establish what constitutes high-quality data. These standards should include requirements for completeness, consistent formatting and naming conventions, standardized classification and categorization, appropriate use of reference values, and expectations for how frequently data is updated to remain current. Validation rules will be applied at point of entry to prevent incomplete or incorrectly formatted data from being stored, such as required fields and restricting acceptable values. ABC should also conduct targeted clean-up efforts to align legacy records with these standards and reduce existing inconsistencies before they continue spreading throughout reporting and operational systems. Once established, data quality will be maintained through shared responsibility across business units which will be supported by ongoing employee training. Monitoring should occur on a routine basis with automated checks identifying anomalies in real time as well as periodic reviews conducted to assess overall data quality and address recurring issues (Stair & Reynolds, 2018).

Although these data quality processes improve accuracy, completeness, and relevance, consistency across systems depends on standardized identifiers and definitions that apply uniformly across the ABC Company's systems. This is achieved through the use of master and reference data governance. Master data represents the core entities of the organization (customer, product, vendor, etc.) while reference data defines the standardized values and classifications used to describe those entities (Gils, 2020). In practical terms, master data acts as the records to be unlocked and reference data functions as the shared key. Both aspects must be consistently defined to ensure that the information retains the same meaning across ABC's systems. Without this structure, data may be accurate within individual systems but may be interpreted differently when used across the organization. These definitions will be developed through collaboration between IT and business data owners as part of the data quality work discussed earlier. This becomes particularly important for ABC due to its interconnected system environment where data is shared across multiple operational and centralized systems. Implementing master and reference data governance will help ABC keep information aligned and improve the reliability of reporting and decision-making as the organization expands to the cloud.

Transparency & Accountability

To support the data quality and consistency efforts, ABC must also establish a strong level of data transparency across internal operations and external stakeholders (customers, vendors, etc.). Transparency ensures that data is not only governed effectively but also clearly understood by those who rely on it. Internally, this means providing employees with easily accessible and understandable data when they have a right to access it (CSU, 2026). Improving transparency in this area reduces the risk of misuse and supports more informed decision-making. For example, employees should be able to trace where data originated, understand how it has been updated across systems, and clearly identify how it is being used in reporting or decision-making. External transparency is also equally important given ABC Company's strong reputation. Transparency is crucial for maintaining trust with their customers and vendors, particularly given the company's reliance on sensitive data. ABC's external stakeholders should

have the ability to ask questions and correct inaccurate information when necessary, ensuring they can understand and engage with how their data is being used. This level of transparency supports trust and responsible data practices by providing stakeholders with meaningful insight into data usage and decision-making processes (Martin, 2022). However, ABC must also recognize that transparency should remain selective. While stakeholders should understand how their data is used, full transparency across all systems and processes may introduce privacy, security, and competitiveness risks if sensitive information is disclosed inappropriately. Transparency efforts should balance accessibility with the organization's responsibility to protect confidential processes and records.

While transparency provides visibility into how data is used and shared, it must also support accountability by clearly defining who is responsible for that data within the company. This means that ABC must establish clear ownership of the data across IT and business units to ensure that specific individuals or teams are responsible for accuracy, integrity, and the appropriate use of data throughout its lifecycle. Data owners within each business unit should be accountable for the quality and correctness of the data they generate, while IT teams are responsible for maintaining the systems and controls that support data governance. This structure ensures that responsibility is not deferred to systems or automation but instead remains clearly assigned to individuals who can address issues when they arise. However, accountability must be distributed appropriately to avoid placing disproportionate responsibility on individuals who may have limited control over system designs or the data processes. In highly integrated environments, there is risk that individuals become "liability sponges" that absorb responsibility for system failures despite not having control over how the data is managed (Martin, 2022). ABC can mitigate this risk by ensuring that accountability is aligned with authority, meaning individuals are only held responsible for areas within their control. As quoted in *Ethics of Data Analytics* (2022), "If the end goal is accountability, then transparency must serve to help locate the various positions of human agency and responsibility in these large and complex sociotechnical assemblages. Ultimately, it is people who must be held accountable for the behavior of algorithmic systems" (p. 404).

Automation Tools

Only after establishing a data management strategy can ABC Company begin implementing automation tools to further support the business. The tools may improve efficiency and scalability, but it can also amplify existing data quality issues when governance practices are weak. Additionally, automation may reduce transparency into how decisions are made and unintentionally reinforce biased outcomes (Martin, 2022). These risks become more significant within interconnected environments when information can spread quickly across systems. ABC must maintain strong governance controls and ongoing, periodic reviews. Human oversight should also remain integrated into automated processes so employees can identify potential issues and maintain accountability. If ABC can maintain these controls, they will see significant ROI such as reducing repetitive manual work, improving consistency across systems, strengthening scalability, and allowing employees to focus more on strategic and higher-value activities.

Master Data Management

Master data management (MDM) will play an important role in supporting ABC Company's automation efforts by helping maintain consistent and reliable information across interconnected systems. MDM establishes a centralized "golden record" that acts as the organization's most accurate and trusted version of business data (Gils, 2020). This is important for ABC because customer, vendor, inventory, financial, and operational data must continuously move across multiple systems to support daily work. Teams across the organization rely on different platforms to complete their responsibilities, yet those systems must work from aligned information to support reporting and decision-making. Given this environment, a coexistence approach is the most appropriate implementation style for ABC. Unlike consolidation, coexistence allows master data to remain synchronized between operational systems and the centralized repository in real time. This is crucial for the company as it allows business units to work primarily from the centralized repository rather than continuing to rely heavily on pulling information directly from operational systems. By maintaining synchronization, employees can rely on the data warehouse as a more trusted and consistently updated source for reporting and decision-making. This will improve operational efficiency by reducing the need for manual verification across systems and reinforce ABC's data management strategy of strengthening the use of the centralized data. Coexistence also supports data consistency and timeliness between systems, improving governance over its data. This style provides stronger scalability and interoperability for ABC's long-term strategy as well. As the company grows, additional systems and data sources can be integrated more effectively without requiring major changes to the overall environment (Robb, 2023). Even though this approach requires greater implementation efforts and infrastructure costs than other MDM styles, the additional investment is justified by ABC's operational complexity and long-term need for synchronized data.

Although coexistence provides significant benefits to ABC Company, there are some negative aspects the organization must consider before deciding on this approach. To begin with, it requires greater implementation complexity, maintenance requirements, and infrastructure costs than simpler MDM approaches. Since the information must continuously synchronize between operational systems and the central repository, ABC will need more advanced integration processes and ongoing system monitoring and governance oversight to maintain reliability (Robb, 2023). Fortunately, many of these implementation efforts can occur alongside the cloud migration strategy. Additionally, one of the largest operational risks associated with coexistence is the reliance on ETL and synchronization processes to maintain records across systems. If data is not transformed or transferred consistently, timing gaps may occur where systems contain poor data quality, similar to quality distortion mentioned prior. Once this data is centralized, it can quickly spread through multiple operational systems. Ultimately, it can reduce trust in reporting and decision-making. Therefore, ABC should implement validation controls, continuous monitoring, synchronization reviews, and routine audits to identify inconsistencies before they spread too far. Strong governance over classifications, transformation rules, and reference values will also remain important to ensure that information maintains the same meaning as it moves between systems (Gils, 2020). Furthermore, maintaining a continuously synchronized central repository increases security exposure as more systems and users will interact with sensitive data. Earlier sections have discussed how interconnected environments can expand the organization's attack surface; coexistence further increases this risk. If ABC continues to implement role-based access controls, encryption standards, periodic access reviews, network segmentation, and continuous security monitoring across systems, they will be able to

keep this exposure much more manageable. Lastly, user adoption remains critical within these types of environments. Even with strong controls in place, inconsistent work practices can quickly create inconsistencies across the environment. Stair and Reynolds (2018) emphasize that long-term process success depends heavily on user understanding and consistent adoption of organizational procedures. This is why ABC must commit to ongoing training to support user buy-in. Although coexistence introduces additional complexity, long-term operational and governance benefits position ABC to more effectively support future automation efforts.

AI & Workflow Automation

Workflow automation can further support ABC Company by reducing repetitive manual processes and improve how information moves throughout the organization. Many routine administrative tasks currently require employee involvement which increases operational efforts, potentially slowing down workflows. Automation can reduce these inefficiencies by automatically routing information between systems through workflow triggers. This will allow ABC employees to spend more time focusing on higher-value strategic initiatives (Murugesan et al., 2023). This provides long-term cost benefits for the organization by reducing unnecessary labor and improving operational speed, resulting in the limited the need for additional staffing as the company continues to grow. Additionally, workflow automation supports ABC's data management strategy by reinforcing standardized documentation practices as it reduces the likelihood of employees bypassing governance procedures during routine tasks and improves consistency across its systems. Since standardized procedures are applied across automated tasks, it may also help reduce certain forms of human bias that can occur during manual decision-making and administrative handling (Upadhyay & Khandelwal, 2018). However, it can still reinforce bias when they rely on poor quality data. This becomes more significant with highly interconnected environments because it can spread quickly to other systems. For example, if outdated classifications or inconsistent documentation standards are introduced into the automated workflow, those issues may continue to be repeated throughout reporting and decision-making. This ties directly back to the governance and consistency concerns discussed earlier in the strategy. To address some of these discrepancies, ABC should continue implementing validation controls, periodic workflow reviews, and continual monitoring to identify inconsistencies before they become a widespread issue. Human oversight must also remain integrated into workflow management so employees can intervene when problems arise and adjust workflows as organizational needs evolve. Additionally, ongoing employee training and governance reinforcement will remain necessary to ensure staff continue following established data management procedures as automation expands throughout the organization.

If implemented selectively and supported by strong governance practices, Artificial Intelligence (AI) may provide additional automation benefits to ABC Company. AI-supported tools can improve efficiency by assisting with large-scale data analysis, identifying patterns within operational data, generating predictive insights, and supporting a more well-informed decision-making process (Alghamdi & Al-Baity, 2022). These capabilities may become increasingly valuable to ABC as they start to manage larger volumes of data across interconnected systems. However, the level of transparency and accountability required within certain departments makes fully automated AI decision-making inappropriate for several areas of the business. Departments such as human resources, vendor

management, customer relations, and other business units responsible for sensitive organizational decisions should avoid relying heavily on AI since they often require contextual judgement and clear accountability. As mentioned prior, organizational trust and governance become more difficult to maintain when employees or external stakeholders cannot clearly understand how decisions are made (Martin, 2022). Furthermore, ABC Company should also be mindful that AI typically reproduces discrimination when the underlying training data reflects past systemic biases. Although ABC has control over data governance within its own systems, machine learning (ML) systems were trained by outside SaaS vendors. This means that bias introduced during model development may still carry into ABC's outcomes. Unfortunately, identifying these biases can also be difficult because organizations do not always understand how recommendations are produced, while many hiring algorithms continue to operate with limited transparency (Bogen & Rieke, 2018). Because of these concerns, ABC should continue following a human-in-the-loop (HITL) approach. Human oversight reduces the likelihood that employees become overly dependent on algorithmic recommendations and allows ABC to identify potential biases or inaccuracies before they affect organizational decisions. HITL combines the efficiency of AI with human oversight which allows employees to intervene when recommendations appear inconsistent or potentially biased (McCormack & Bendeckache, 2024). "By maintaining human oversight at key stages, the organization can reduce the risk of blindly trusting algorithmic outcomes while still benefiting from AI's strengths. Responsible hybrid models offer more flexibility and nuance than fully automated systems" (Schwenk, 2025, para. 12). With that being said, HITL introduces additional labor involvement and reduces some of the speed advantages associated with fully automated AI systems. It also increases operational costs as employees must remain actively involved with decision-making. Additional investments in employee training, governance oversight, continual monitoring, and periodic system audits will be required as well. For ABC, the additional oversight is still worthwhile because it aligns more closely with the organization's broader governance and accountability goals.

Business Intelligence & Analytics

Even though automation improves efficiency and the potential for consistency, ABC Company needs a plan to ensure that their data management strategy continues to operate effectively. Because of the complexity of ABC's data environment, leadership must be able to evaluate how effectively data is being managed across its systems. Business intelligence (BI) and analytics will provide the organization with the visibility necessary to monitor governance compliance and security risks. They can support these efforts by transforming data into meaningful reporting structures that help track data quality issues, access activities, workflow inconsistencies, synchronization failures, and other governance concerns outlined earlier in this proposal. More importantly, these tools allow ABC to identify trends that can be corrected and investigate the root cause of reoccurring issues before problems spread throughout their systems. BI therefore gives leadership another way to identify governance and operational concerns once the strategy is implemented. However, maintaining governance in a BI environment will require significant investment by ABC Company. They will need personnel responsible for oversight, maintenance, analytics development, and ongoing monitoring to ensure reports remain accurate and aligned with company standards. Infrastructure costs and SaaS agreements associated with analytics software must be considered as well. Governance Platforms can range from \$50,000 to

\$150,000 per year depending on data volume, number of users, deployment model compliance/security needs, and automation maturity (Shaikh, 2026). Despite this, many organizations continue to heavily invest in analytics because long-term visibility, governance, and decision-making are positively affected. For ABC, this is just a small price to pay considering the average cost of a data breach in the U.S. is \$4.4 million (Shaikh, 2026).

Metadata Management

ABC Company must maintain strong metadata (“data about data”) management practices across its reporting environment in order to effectively support governance oversight. Metadata provides important contextual information such as where the data originated, how it was transformed, who is responsible for it, how it is classified, and how it is being used (Marco, 2026). Within ABC’s environment, metadata will support many of the data management objectives discussed earlier in this strategy, especially transparency, accountability, classification, and the ETL process. This gives governance teams a more reliable way to trace inconsistencies and identify where governance standards are not being followed. When it comes to storing this metadata, the organization should implement a distributed metadata architecture that aligns with the data marts that already exist within its data warehouse structure (Figure 1). Under this structure, metadata repositories may remain segmented by operational functions while still supporting enterprise-wide governance oversight. This will limit unnecessary visibility into sensitive information which is important for privacy and security. For example, HR personnel should not have unrestricted access to finance metadata and vice versa. Partitioned metadata coupled with the data marts themselves helps reinforce role-based access controls and supports the governance principles discussed earlier. ABC will need to be mindful that in this type of architecture costs can be higher and control of the metadata can diverge between local solutions (Gils, 2020). Ongoing governance oversight and employee training will still be necessary to keep standards and classifications aligned across the different marts. This gives ABC a more reliable way to organize and track information before it is ultimately used within dashboards and reporting.

Dashboards & Reporting

By maintaining a centralized data warehouse and metadata registry, ABC can access and combine metadata even if they are distributed across data marts. This allows the company to pull governed, standardized information into a centralized BI environment rather than relying heavily on fragmented systems for analytics and reporting. Within the BI tool, dashboards and reports can be used to support ongoing governance monitoring previously mentioned throughout the proposal. ABC should implement reports and dashboards that track data quality trends, synchronization failures, workflow inconsistencies, access activity, and other operational issues that may indicate weakness described throughout the data management strategy. Security and compliance dashboards should also be utilized to monitor unusual access behavior that may signal security concerns. Automated alerts and workflow triggers may be incorporated into these monitoring environments to notify governance or security teams when unusual activity is detected. However, ABC should continue to follow the HITL approach as these systems may produce false positives for normal operational behavior. Leadership can also use these reporting tools to identify recurring issues and determine whether corrective actions are actually improving operations over time. These reports should be reviewed routinely to ensure recurring issues

are identified within a timely manner. Using a centralized reporting tool extends beyond monitoring of governance and security. When end users work from the standardized data sources instead of developing disconnected reports from operational systems, it reduces the likelihood of conflicting metrics, unintended biases, untimely reporting issues, and variations on how the information is interpreted. Fortunately, ABC already utilized Power BI, and this is a matter of optimization vs. full system implementation.

Ethical Principles

Ethical principles have been heavily incorporated throughout ABC Company's data management and security strategy. Earlier sections discussed the importance of transparency, accountability, privacy, governance oversight, and maintaining consistent data standards across the organization. These concepts work together to help ensure that information is managed responsibly and reduce the risk associated with poor governance practices. Controls such as role-based access, metadata management, validation processes, centralized reporting, and ongoing monitoring are not only important for security and consistency, but also for ensuring the data remains valid, accurate, complete, and relevant. The strategy also recognizes that ethical governance requires balance. Although transparency is important for accountability and trust, ABC must protect sensitive information from unnecessary exposure. They need to make sure that the data is being used by the right people and for the right reasons. These ethical considerations will become increasingly important as ABC continues to scale and migrate into the cloud environment. Without strong governance practices in place, concerns may continue and spread quickly throughout the company system's environment.

ABC Company will also need to consider the ethical considerations while implementing automation, AI tools, and analytics across the organization. Although these can improve efficiency and strengthen governance, these tools may also introduce risks if they are not governed effectively. They can reinforce bias if the underlying training data reflects historical discriminatory patterns. ABC may struggle to fully understand how algorithmic systems generate recommendations, creating concerns surrounding transparency and accountability. Because of this, it is especially important for the company to be selective with its automation and AI efforts when it can influence sensitive decisions involving employees, customers, or vendors. Even when processes do warrant automation, ABC's strategy emphasizes HITL. However, they must be careful that employees do not become "liability sponges" if they only have limited control over how the systems are designed or trained. These controls help ensure that automation continues supporting organizational decision-making without removing accountability from employees and leadership.

Lastly, it is ABC Company's responsibility to consider how the organization represents itself publicly from an ethical viewpoint. As the company continues collecting and managing large volumes of information, they must ensure that the data is being used in ways that align with their values and customer expectations. Customers and vendors should have confidence that their information is not being misused, shared, or utilized in a way that conflicts with the purpose in which it was originally provided (Martin, 2022). This is not only for legal compliance, but to ensure that the ethical usage of stakeholders' data would not be viewed as overly intrusive. Even if certain data practices are technically

permitted, they still might damage trust. Because ABC is known for having a strong reputation for quality and customer relationships, unethical practices could create long-term consequences. Therefore, they must ensure that policies, standards, vendor agreements, and communications practices reinforce responsible data usage across the organization. This will better protect stakeholders (internal and external) confidence and support long-term reputational sustainability.

Final Thoughts

As ABC Company continues expanding its cloud environment, long-term success will depend on how effectively the organization manages security, governance, scalability, operational efficiency, and ethical responsibility together. Although this strategy introduces significant complexity and costs, the investment is necessary to reduce larger financial and reputational risks associated with poor data management practices. Many of these proposed improvements can be implemented alongside the cloud migration strategy which will allow ABC to strengthen capabilities without creating unnecessary operational disruption. Spreading implementation across phases over the next 12–18 months will give ABC more flexibility to prioritize higher-risk systems and governance improvements before moving onto the broader environment. Though this might be a long and expensive road, it positions ABC Company to support future growth with greater confidence in their systems, governance practices, and decisions that drive the business forward.

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AI Use Statement

ChatGPT (OpenAI) was used to support brainstorming, sentence refinement, paragraph structure, and alignment of prior coursework with current assignment requirements. Prompts focused on strengthening transitions between sections, improving clarity of written explanations, evaluating where previous academic work could support new rubric requirements, and refining source integration. All analysis, interpretation, conclusions, and final written decisions remain the author's own work.